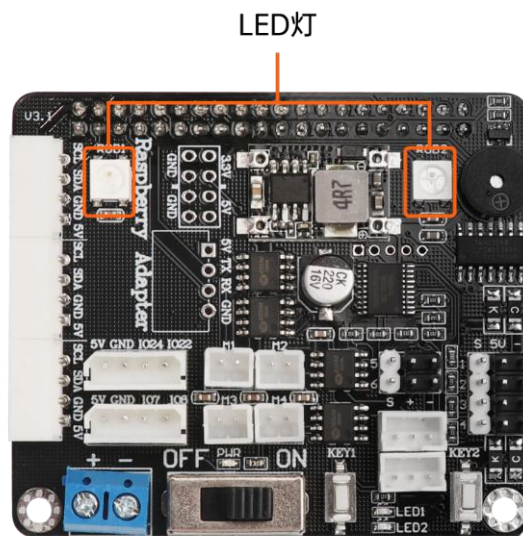


Lesson 3 Control RGB Light

1. Preparation

There are two RGB lights on the Raspberry Pi expansion board, as shown in the figure below:



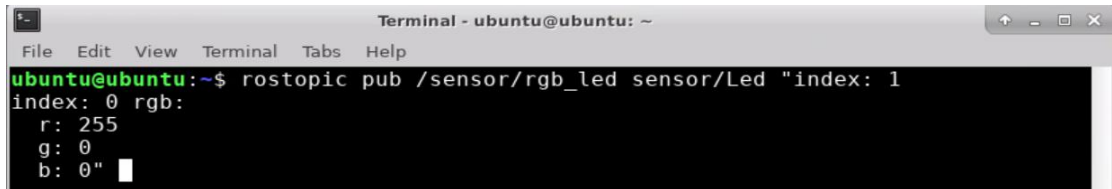
2. Operation Steps

1) Open LX terminal. Click  Applications and select  Terminal Emulator to open command line terminal of Ubuntu.

2) Initially, enter “**rostopic pub /sensor/rgb_led sensor/Led "index: 0**” command and complete the command through TAB key.

```
Terminal - ubuntu@ubuntu: ~
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~$ rostopic pub /sensor/rgb_led sensor/Led "index: 0
rgb:
  r: 0
  g: 0
  b: 0"
```

3) Next, revise the parameter of the “index” and “rgb” in the command. For example, we modify the parameter of index as 1 and rgb as r: 255 g: 0 b: 0. Then press Enter.

A screenshot of a terminal window titled "Terminal - ubuntu@ubuntu: ~". The terminal shows a command being executed: `rostopic pub /sensor/rgb_led sensor/Led "index: 1`. The output of the command is displayed on the next line: `index: 0 rgb:`, followed by `r: 255`, `g: 0`, and `b: 0"`. The cursor is positioned at the end of the output line.

```
Terminal - ubuntu@ubuntu: ~
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~$ rostopic pub /sensor/rgb_led sensor/Led "index: 1
index: 0 rgb:
r: 255
g: 0
b: 0"
```

Note: when the input parameter of “index” is 0, the object to be controlled in the program is RGB Light 0. When is 1, the object is RGB Light 2. “rgb” refers to the color of RGB light.

4) If you want to quit the program, please press “Ctrl+C”.

3. Project Outcome

After the program running, RGB light 2 on the raspberry Pi extension board will flash in the color we have set before.